



Enhanced Mapping System - Implementation Guide

What's New

🌟 Features Added

1. **Automatic Geocoding** - No more manual coordinates!
 2. **Hierarchical Location System** - Barangay → Sitio → Individual learners
 3. **Marker Clustering** - Better visualization with many pins
 4. **Precise Pins** - Exact coordinates from addresses instead of barangay centers
 5. **Batch Geocoding Tool** - Update existing learner coordinates automatically
-



Implementation Steps

Step 1: Install Required Python Package

```
bash
pip install httpx
```

Step 2: Update Your Database Schema

Add these columns to your `als_afl` table:

```
sql
ALTER TABLE als_afl
ADD COLUMN house_number VARCHAR(50),
ADD COLUMN street VARCHAR(100),
ADD COLUMN sitio_purok VARCHAR(100),
ADD COLUMN geocoded BOOLEAN DEFAULT FALSE;

-- If lat/lng don't exist:
ALTER TABLE als_afl
ADD COLUMN lat DECIMAL(10, 8),
ADD COLUMN lng DECIMAL(11, 8);
```

Step 3: Replace Backend Code

In your `ALS.py`, add the enhanced backend code from artifact `enhanced_backend`.

Key additions:

- `geocode_address()` function

- Enhanced AF1 learner model
- New endpoints:
 - `POST /api/af1-learners` - Create with auto-geocoding
 - `POST /api/af1-learners/batch-geocode` - Batch geocode existing
 - `GET /api/af1-learners/by-location` - Hierarchical data

Step 4: Replace Frontend JavaScript

Replace your `instructor_mapping.js` with the enhanced version from artifact `enhanced_mapping.js`.

Key features:

- Marker clustering support
- Hierarchical location tree
- Sitio/barangay markers
- Toggle to show/hide sitios

Step 5: Update HTML Files

1. Update `instructor_mapping.html`:

- Add MarkerCluster library (see artifact)
- Links already provided in enhanced HTML

2. Replace `instructor_af1_data_entry.html`:

- Use the enhanced form from artifact `enhanced_af1_form`
- Includes address fields and auto-geocoding feedback

3. Add new utility page:

- Save `batch_geocode_ui` as `HTML/batch_geocode.html`
- Add route in `ALS.py`:

```
python

@app.get("/batch_geocode", response_class=FileResponse)
async def batch_geocode_page():
    return FileResponse("HTML/batch_geocode.html")
```

Step 6: Update CSS

Add the cluster styles from artifact `cluster_css` to your `instructor_mapping.css`.



Address to Coordinates Flow

User enters:

— House Number: "123"

— Street: "Main Street"

— Sitio: "Sitio 1"

— Barangay: "Alipit"

Backend combines into:

"123 Main Street, Sitio 1, Barangay Alipit, Santa Cruz, Laguna, Philippines"

Geocoding API returns:

lat: 14.2299123, lng: 121.4102456

Saved with geocoded: true

Fallback System

If geocoding fails:

1. Uses barangay center coordinates
2. Sets `geocoded: false`
3. Learner still shows on map (approximate location)



Usage Guide

For New Learners

1. Go to AF1 Data Entry form
2. Fill in complete address:
 - House number (optional but recommended)
 - Street name
 - Sitio/Purok **required**
 - Barangay **required**
3. Submit form
4. System automatically:
 - Geocodes address
 - Saves precise coordinates
 - Shows success with geocoding status

For Existing Learners

1. Navigate to `/batch_geocode` page
 2. View statistics:
 - Total learners
 - Already geocoded
 - Need geocoding
 3. Click "Start Batch Geocoding"
 4. Wait for completion (1 learner/second)
 5. View updated map with precise locations
-



Map Features

Visualization Types

1. **Barangay Pins** (Blue)
 - Shows total learners per barangay
 - Positioned at barangay center
2. **Sitio Pins** (Yellow/Orange)
 - Shows learners per sitio
 - Positioned at sitio center
 - Toggle on/off with button
3. **Clustered View**
 - When zoomed out, nearby pins cluster
 - Shows count in cluster
 - Click to zoom and expand

Interactive Elements

- Click pin → See learner details
 - Click barangay in sidebar → Zoom to location
 - Toggle sitios on/off
 - View geocoding status for each learner
-



Configuration

Geocoding Service

Currently uses **Nominatim (OpenStreetMap)**:

- Free and open source
- Rate limit: 1 request/second
- No API key required

To use a different service (Google Maps, Mapbox), modify the `geocode_address()` function.

Marker Cluster Settings

In `instructor_mapping.js`:

```
javascript

clusterGroup = L.markerClusterGroup({
  maxClusterRadius: 50, // Adjust clustering distance
  spiderfyOnMaxZoom: true, // Expand clusters on max zoom
  showCoverageOnHover: false // Hide cluster coverage area
});
```

Troubleshooting

Issue: Geocoding not working

Check:

1. Internet connection
2. Address format (must include barangay)
3. Rate limiting (wait 1 second between requests)

Solution: Use fallback coordinates temporarily

Issue: Pins not showing on map

Check:

1. Browser console for errors
2. MarkerCluster library loaded
3. Learners have lat/lng values

Solution: Run batch geocode for existing learners

Issue: Coordinates inaccurate

Cause: Geocoding service doesn't recognize exact address

Solution:

1. Add more address details (street, house number)
 2. Manually verify important locations
 3. Use different geocoding service
-



Performance Tips

1. **Batch Process:** Use batch geocoding during off-hours
 2. **Cache Results:** Geocoded coordinates are permanent
 3. **Progressive Loading:** Map loads in chunks
 4. **Clustering:** Reduces marker count for better performance
-



Customization

Change Pin Colors

In `instructor_mapping.js`:

```
javascript

const color = '#667eea'; // Barangay color
const sitioColor = '#FFC107'; // Sitio color
```

Adjust Map Center

```
javascript

map = L.map('map').setView([14.2818099, 121.414977], 13);
//           ↑ Latitude ↑ Longitude ↑ Zoom
```

Modify Clustering Distance

```
javascript

maxClusterRadius: 50 // Pixels - larger = more clustering
```



Security Notes

1. **Rate Limiting:** Geocoding respects 1 req/sec limit
2. **Input Validation:** All addresses sanitized before geocoding

3. **Fallback:** Always provides coordinates (precise or approximate)

Support

If you encounter issues:

1. Check browser console for errors
 2. Verify database schema matches requirements
 3. Ensure all dependencies installed
 4. Test with a single learner first
-

Benefits

- ✓ No more manual coordinate entry
- ✓ Precise learner locations
- ✓ Better enrollment planning
- ✓ Visual sitio-level insights
- ✓ Scalable to thousands of learners
- ✓ Works offline after initial geocoding

Enjoy your enhanced mapping system!  